Self-Hosted LLMs for Accessible Software Engineering: A Drift-, Bias- and Cost-Aware Evaluation on an 8 GB Consumer GPU

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***Abstract*—LLMs now write and repair user-interface code, yet LLM comparisons rarely measure accessibility, mostly test one-shot prompts, sometimes let a model judge its own output, and seldom report cost. In our framework, six WCAG 2.2-anchored capability dimensions are scored by real-browser oracles (axe-core and 27 keyboard and semantics probes). The *Dynamic Contextual Drift Index* (DCDI) measures the accessibility adherence an agent loses over a long, tool-using session, against fresh-context and length-matched padded controls. *Cross-Model Blind Peer Evaluation* (CMBPE) is an anonymised, family-recused, order-permuted and length-adjusted jury for alt text, calibrated against ratings by blind and low-vision (BLV) people. *Capability-per-Dollar-Second* (CPDS) relates capability to cost and latency. We apply the framework to four 4-bit open-weight models that run for free on one 8 GB consumer GPU (Qwen3 8B, Qwen2.5-VL 7B, Gemma 3 4B, Ministral 3 8B), with 10,937 answered calls. Asked for accessibility, Qwen3 8B generated pages that passed most checks ($S_{\text{gen}} = 0.84$), but only a share of 0.15 passed all, and the request made both E1 models far more likely to claim conformance for failing pages. In repair and audit, Qwen3 8B scored 0.36 and 0.34 (F1), Qwen2.5-VL 7B 0.04 and 0.00. In a 24-step agentic session, DCDI detected no drift (-0.05), yet in the second half Qwen3 8B ended 10 of 12 steps without touching a file while reporting the change as done; DCDI drops such steps. CMBPE beat the median single judge on known-answer alt-text pairs but tracked BLV ratings no better than description length. Self-hosted repair costs no money but about 74 GPU hours per 1,000 pages. All values are computed by released code from logged responses.**

***Index Terms*—accessibility, WCAG 2.2, large language models, LLM-as-a-judge, agents, evaluation, alt text**

I. Introduction

Most web pages still fail automated accessibility checks, and code assistants now write a growing share of user-interface code. Whether that code is accessible therefore matters for blind, low-vision, motor-impaired and keyboard-only users. Studies of LLMs and accessibility report that generated code often misses accessibility requirements unless they are requested, and that models can fix many violations once they are pointed out [1], [2]. These studies usually test one or two models with one-shot prompts, judge outputs with a single (sometimes the same) model, and do not report what an evaluation or a remediation costs.

Four practical questions remain open. Does a model build accessible interfaces *by default*, or only when told to? Does it keep honouring accessibility while it works through a long, tool-using session, where its context fills with history [3], [4]? Can LLM judges rate accessibility content such as alt text reliably, given their known position, length and self-preference biases [5], [6], [7]? And what does AI-assisted remediation cost per page, in money and in time?

We answer these questions with a framework built around automated, browser-based oracles and three new measures (DCDI, CMBPE and CPDS), and we apply it to the setting most relevant to organisations without an API budget: small open-weight models served locally on one 8 GB consumer GPU. Our research questions are:

- **RQ1** How do the models perform across six accessibility-engineering capability dimensions (C1–C6)?
- **RQ2** Do LLM agents keep honouring accessibility requirements as a long, tool-using session grows?
- **RQ3** Does a blind, family-recused, order-permuted, length-adjusted jury (CMBPE) judge accessibility content more reliably than single LLM judges?
- **RQ4** Which models are Pareto-efficient in capability, cost and latency, and what does AI-assisted remediation cost?

Contributions. (1) A capability taxonomy that maps general LLM abilities to six WCAG 2.2-anchored tasks with automated oracles, including 27 keyboard and semantics probes, each validated on a passing and a failing fixture. (2) DCDI, a drift measure with a counterfactual fresh-context control and a padded control that separates the effect of context length from the effect of the session's history. (3) CMBPE, a bias-controlled jury with a Bradley–Terry model of position and length bias. (4) CPDS, a scenario-weighted efficiency score with bootstrap Pareto membership. (5) A fully logged study of four self-hosted models, with code, prompts, raw scores and call logs released.

Scope. This study evaluates 4–8B open-weight models at 4-bit quantisation, served by llama.cpp [8] on one consumer GPU. It does not evaluate frontier closed models. The harness supports them, but running them needs paid API access. Claims about the frontier, or about closed versus open models, are therefore out of scope here.

II. Related Work

LLMs and accessible code. Aljedaani et al. studied the accessibility of web code generated by ChatGPT and its ability to repair violations [1]. CodeA11y adds accessibility-aware suggestions and reminders to an AI coding assistant and evaluates them with developers [2]. We complement such studies with repeated, multi-model measurement, a repair corpus with known faults, long agentic sessions and cost.

Automated accessibility testing. Rule engines such as axe-core [9] check a subset of WCAG success criteria [10] in a rendered page; keyboard operability, focus management and widget state largely escape them. We pair axe-core with browser-driven probes (Playwright [11]) and treat all automated results as lower bounds on failures.

LLM-as-a-judge. LLM judges favour the first-shown or longer answer and their own outputs [5], [6], [7]. CMBPE combines several known mitigations (both orders, anonymisation, recusal) with an explicit Bradley–Terry model [12] of position and length bias, and calibrates the jury against human ratings.

Long contexts and agents. Models use information in long contexts unevenly [3], and effective context is often shorter than the advertised window [4]. Agentic coding benchmarks measure task success [13] but not whether non-functional requirements stated once at the start are still honoured many steps later. DCDI measures exactly that.

Alt-text evaluation. Kreiss et al. showed that context matters for the usefulness of image descriptions to blind and low-vision (BLV) readers and collected BLV ratings [14]. ContextRef provides crowd ratings of descriptions in Wikipedia context [15]. We use both as calibration data for LLM judges.

III. Framework

A. Oracles and the rule-level pass rate

Pages are rendered in headless Chromium 156.0.8078.4 at 1280×800 and served from a local HTTP server that blocks every other host. axe-core 4.14.0 runs with the WCAG 2.x Level A and AA rule tags. Pages with timers are checked again after 6 s, and a rule fails if it failed in either run. 27 probes cover what axe cannot see. Three generic probes run on every page: tab reachability, visible focus and no keyboard trap. The others cover widget patterns: form errors associated with their fields and not shown by colour alone, dialog focus handling, tabs, accordions, disclosure navigation, comboboxes, sliders, switches, tooltips, live regions, data-table headers, sort controls, pause controls for moving content, multi-step focus management, captions, landmarks, headings, language of parts, contrast after a theme toggle and accessible names that are not placeholder-only. Every probe passes its accessible fixture and fails its inaccessible fixture. Probes find widgets by visible text anchors, never by implementation details, and accept any keyboard path that satisfies the success criterion.

TABLE I
Capability dimensions, experiments and scores.

Dimension	Experiment	Score
C1 conformance auditing	E1-Audit	S_1 = micro-F1 on 192 known faults
C2 accessible UI generation and repair	E1-Gen, E1-Repair	$S_2 = (S_{\text{gen}} + S_{\text{rep}})/2$
C3 agentic execution of a multi-page build	E2	S_3 = mean in-session adherence
C4 text alternatives for images	E3c	S_4 = CMBPE expected win rate
C5 adherence as context grows	E2	$S_5 = 1 - \text{clip}(\text{DCDI}, 0, 1)$
C6 conflicting requests and honest claims	E2, E1-Gen	S_6 = (pressure score + (1−overclaim rate))/2

For an artefact x, let $Q(x)$ be the applicable checks: every axe rule that evaluated at least one node in scope, plus every applicable probe. The pass rate is

$$a(x) = |\{q \in Q(x) : q \text{ passes}\}| \,/\, |Q(x)|. \tag{1}$$

Task-specific functional checks (for example "an empty submit is blocked") confirm that the page does what was asked.

B. Capability taxonomy

Table I lists the six dimensions and their scores $S_k \in [0, 1]$. The composite capability score is the weighted geometric mean $\text{CCS}_m = \exp\big(\sum_k w_k \ln \max(S_{m,k}, 0.01)\big)$ with $w_k = 1/6$, reported only for models with all six scores; a geometric mean keeps a weak dimension from being masked by strong ones.

C. Dynamic Contextual Drift Index (DCDI)

An agent works through a fixed sequence of steps on one workspace. After each step t of run r we score the in-session adherence $a_{r,t}$ on the step's scope and checks. A *fresh control* replays the same step in a new conversation on a copy of the workspace snapshot $S_{r,t-1}$, with the same system prompt and tool budget, giving $a^0_{r,t}$. Over the foundation and regular steps T_{reg} that are functional in both arms,

$$\text{DCDI}_m = \frac{\sum_{r,t \in T_{\text{reg}}} (a^0_{r,t} - a_{r,t})}{\sum_{r,t \in T_{\text{reg}}} a^0_{r,t}}. \tag{2}$$

DCDI is 0 without drift, 1 when all adherence is lost and negative when history helps. A *padded control* at checkpoints K replays the step after irrelevant history whose length matches the session's prompt length $\Lambda_{r,t}$ within $\pm 10\,\%$. It splits the drift at the checkpoints into a length and a history component, $\text{DCDI}_K = [\sum(a^0 - a^{\text{pad}}) + \sum(a^{\text{pad}} - a)]/\sum a^0 = \text{DCDI}_{\text{len}} + \text{DCDI}_{\text{hist}}$. The drift rate β_m is the OLS slope of $a^0 - a$ on $\log_2(\Lambda/8192)$ with step fixed effects (adherence lost per doubling of context).

D. Cross-Model Blind Peer Evaluation (CMBPE)

Judges from different model families compare two anonymised candidates for the same image in both orders.

Candidate texts are stripped of quotes, markdown and lead-ins such as "Image of", then given random opaque ids. A verdict is dropped from aggregation when the judge's family equals either candidate's family (recusal). With $y = 1$ if the first-shown candidate wins,

$$P(y = 1) = \sigma(\theta_a - \theta_b + \delta_j + \gamma_j \, \Delta\ell), \tag{3}$$

where $\Delta\ell$ is the standardised length difference, δ_j and γ_j are judge j's position and length biases, and $\sum \theta = 0$. The model is fitted by penalised maximum likelihood (ridge 10^{-3} on θ) with image-cluster bootstrap intervals. A generator's score is its expected win rate $S_4(m) = \mathrm{mean}_{g \neq m} \, \sigma(\theta_m - \theta_g)$. Self-preference η_j is estimated from judge j's own-family verdicts with θ fixed at the panel estimate. Position consistency PC_j is the share of pairs with the same verdict in both orders. Baselines are a single judge (one seeded order, no recusal or adjustment), a naive majority panel and a length-only scorer.

E. Capability-per-Dollar-Second (CPDS)

For model m and task family f (nudged generation, repair), with capability score CS, mean billed cost per task C and median latency L,

$$\mathrm{CPDS}^{(\alpha,\beta)} = \frac{CS}{\phi(C/c_0)^{\alpha} \, \phi(L/l_0)^{\beta}}, \quad \phi(z) = 1 + \log_{10}(1+z), \tag{4}$$

with $c_0 = \$0.01$ and $l_0 = 1\,\mathrm{s}$. The scenarios are batch remediation ($\alpha = 1, \beta = 0$, on repair), interactive assistant ($\alpha = 0.5, \beta = 1$, on generation) and balanced ($\alpha = \beta = 1$). A model's Pareto-membership probability π_m is the share of B bootstrap resamples in which no other model is at least as capable, cheap and fast and strictly better in one of them. Self-hosted models have $C = 0$ (electricity is not metered), so CPDS reduces to a latency-adjusted capability on the stated hardware.

IV. Study Design

A. Models, hardware and serving

Table III lists the models. Qwen3 8B [16] is a text model with thinking on by default; it runs E1 and E2. Qwen2.5-VL 7B [17], Gemma 3 4B [18] and Ministral 3 8B [19] are vision-language models from three families and form the E3 generator pool and jury; Qwen2.5-VL 7B also runs E1. All weights are the publishers' Q4_K_M GGUF files, pinned by SHA-256. They are served by llama.cpp b12004 [8] through its Vulkan backend on the open Mesa NVK driver, on an NVIDIA RTX 2060 SUPER (8 GB), an Intel i7-9700K and 32 GB RAM (Ubuntu 26.04). Every layer is on the GPU and one model is resident at a time. Qwen3 8B uses its native 32,768-token window with an 8-bit KV cache and may emit up to 32,768 tokens, as its model card recommends. The other models may emit up to 8,192. The harness sends no sampling parameters. The server applies each model's Ollama-library defaults (Qwen3 8B: temperature 0.6, top-k 20, top-p 0.95; Qwen2.5-VL 7B: 0.0001; Gemma 3 4B: 1.0, 64, 0.95; Ministral 3 8B: 0.15), so Qwen2.5-VL 7B is close to greedy. Images are encoded with at most 1,280 visual tokens (Gemma 3 4B uses its fixed 256). A prompt longer than the window is rejected by the server and recorded as context exhaustion, never truncated. Requests are serialised, so latencies are those of one request on this machine. Table II summarises the serving speed and GPU memory measured during the study (`experiments/local/extra_values.py`). The desktop session shares the GPU, so the memory figures include about 1.3 GB that the study does not use.

TABLE II
MEASURED SERVING SPEED AND GPU MEMORY (MEDIAN OVER STUDY CALLS; VRAM OVER 10 S SAMPLES WHILE THE MODEL WAS WORKING; ALL PROCESSES).

Model	Decode tok/s	Prompt tok/s	VRAM MiB
Qwen3 8B	49.2	1213	8118
Qwen2.5-VL 7B	32.0	270	6859
Gemma 3 4B	47.0	276	4619
Ministral 3 8B	24.7	105	6387

TABLE III
MODELS ACTUALLY RUN (GENERATED FROM `CONFIG/MODELS.YAML`). V: VISION. TTC: TEST-TIME COMPUTE. USE: 1 = E1, 2 = E2, G = E3 GENERATOR, J = E3 JUDGE. SELF-HOSTED MODELS HAVE NO PER-TOKEN PRICE.

Model	Open	Arch.	V	TTC	$ in/out	Use
Qwen3 8B (local)	✓	dense 8B	–	thinking	0/0	1 2
Qwen2.5-VL 7B (local)	✓	vision-language 7B	✓	–	0/0	1 G J
Gemma 3 4B (local)	✓	dense 4B	✓	–	0/0	G J
Ministral 3 8B (local)	✓	dense 8B	✓	–	0/0	G J

B. E1: generation, repair and audit (C1, C2, C6)

Generation. 20 interface tasks: form with validation, modal dialog, tabs, accordion, disclosure navigation, data table, image gallery with lightbox, auto-rotating carousel, autocomplete combobox, switch, slider, status message, article structure, video, pagination, tooltip, file upload, multi-step form, consent banner and icon links. Each runs *unguided* and *nudged* (one added sentence: "Make sure it is accessible and conforms to WCAG 2.2 Level AA."), with 3 samples each. The model must return one self-contained HTML document; a missing or truncated block scores 0. A second turn asks whether the page conforms to WCAG 2.2 AA (JSON). S_{gen} is the mean over nudged samples of $a(x) \cdot \mathbb{1}[\text{functional}]$. The *default gap* is $S_{\text{gen}}(\text{nudged}) - S_{\text{gen}}(\text{unguided})$. The *overclaim rate* is the share of functional, imperfect pages ($a(x) < 1$) that the model claims conform; it is a lower bound, because automated checks miss failures.

Repair. 12 accessible base pages: 4 adapted W3C "Before and After" demonstration pages [20] and 8 authored pages. 12 mutation operators inject faults: image alt removed (SC 1.1.1), label replaced by a placeholder, contrast reduced, icon-button name removed, icon-link name removed, page language removed, button turned into a `div`, heading demoted, positive `tabindex`, focus indicator removed, invalid ARIA attribute

and page title removed. A seeded search yields 48 faulty pages with 4 faults each (192 faults, 16 per operator), and every injected fault is detected by the oracles. A repair scores $s = \max(0, (\text{resolved} - \text{regressions})/4) \cdot \mathbb{1}[\text{preserved}]$, where preservation requires visible-text similarity ≥ 0.95, every base-page id and DOM tag-sequence similarity ≥ 0.85. S_{rep} is the mean page score.

Audit. The model lists the problems of each faulty page as JSON. Reports are matched one-to-one to injected faults by success criterion and element. S_{audit} is the micro-F1 over the 192 faults; unmatched reports count as false positives, a lower bound on precision.

C. E2: long agentic sessions (C3, C5, C6)

The model builds a small clinic website with three file tools (`list_files`, `read_file`, `write_file`) in a sandboxed workspace. The system prompt states the WCAG 2.2 AA requirement once; no user message mentions accessibility. There are 24 steps: 4 foundation pages (services, appointments, contact, news), then 14 regular features (hero banner, site search, accordion, form validation, confirmation dialog, cookie banner, gallery, language switcher, pagination, back-to-top button, social links, dark-mode switch, news ticker, pain-level slider) and 6 *pressure* requests that conflict with accessibility (placeholder-only labels, light-grey footer text, a click-only tile, removing focus outlines, colour-only validation errors, disabling zoom), in a fixed random order. A step ends when the model answers without a tool call or after 10 tool calls; history is never truncated or summarised. A pressure step scores 1 if the page passes all its checks afterwards and, for requests that can be met accessibly, the request was met; 0.5 if such a request was not met but the page passes; and 0 if the page fails a check afterwards, for whatever reason. Fresh controls run at every step, padded controls at positions 8, 12, 16, 20 and 24 (only where the session reaches them), and an end-of-session audit re-scores every earlier step.

D. E3: alt text and judges (C4)

D_{BLV}: 68 descriptions in article context, each rated by four BLV participants [14]. D_{CR}: 204 descriptions in Wikipedia context with crowd ratings [15]. Images are the datasets' own stimuli (222 distinct images, all retrieved). The judges, one per family, rate every item on a 5-point scale (pointwise) and compare pairs in both orders. *E3a* pairs two D_{BLV} descriptions of the same image whose BLV ratings differ by at least 0.5 (23 pairs). *E3b* pairs each original description with four degraded versions (another item's first sentence appended, repeated text, another image's description, the first three words), giving 1,058 known-answer pairs. *E3c* has each generator write alt text for 120 seeded D_{CR} images. With three generators all three pairs per image are judged, and after recusal each pair keeps exactly one judge, from the third family.

E. Statistics

We report 95 % percentile bootstrap intervals ($B = 2000$, seed 20271113) that resample the natural cluster: task (E1-Gen), page (E1-Repair, E1-Audit), run $\times$ step (E2) and image (E3) [21]. All models share the same resamples, so differences are paired. We call two models different only when the interval of their difference excludes 0. Hypothesis tests are one-sided bootstrap tests with Holm correction [22]. Inter-judge agreement is Krippendorff's α [23].

F. Pre-registered hypotheses and their status in this study

The framework's pre-registered hypotheses are: H1, larger siblings drift less than smaller ones (vendor pairs); H2, the best closed model's C4 score exceeds the best open model's; H3, at least one open-weight model has $\pi \geq 0.5$ on the batch-remediation frontier; H4, (a) some single judge prefers its own family and (b) CMBPE beats the median single judge on known-answer pairs; H5, for most models nudged generation beats unguided generation. With four local open-weight models, H1 and H2 cannot be tested and H3 holds trivially. H4 and H5 are tested as planned.

G. Changes from the original plan

The original plan targeted frontier and hosted models through paid APIs. Without an API budget, this study runs the plan's documented local profile, with these changes (all logged, dated and motivated in `experiments/DEVIATIONS.md` before the affected calls): llama.cpp instead of Ollama, because the latter has no GPU backend on the open driver; a third local judge family (Ministral 3 8B), without which recusal removes every E3c verdict; a 32,768-token output budget for Qwen3 8B (the plan's rule: the model's maximum output); the image-token cap; the E2 padding transcript written by Qwen3 8B instead of Claude Haiku 5.5, kept to the turns that fit its window; and one E2 run (seed 101) instead of three. No metric, prompt, task or scoring rule was changed. During the run we fixed the harness and the run script without changing what is measured. A response that fills Qwen3 8B's window is now recorded as context exhaustion, because llama.cpp signals a full window by stopping the output rather than by rejecting the request. A padded control runs only after a completed session step, and tool-call arguments that are not valid JSON are sent back to the model as an empty object. Leakage-check folds are grouped by image. The pipeline now checks which server profile is loaded and that the vision encoder works (see Threats to Validity). E2 run 0 is the pilot (same seed and model), so its first six positions reuse the pilot's cached responses. Two pairs of repair and audit pages turned out to be identical (B02-V1/V2 and B03-V1/V2), leaving 46 distinct pages with 184 distinct faults. We kept the frozen corpus and also report E1 without the duplicates. Every page is rechecked after 6 s, not only pages with timers, which is equal to or stricter than the plan. After seeing the E2 data we added exploratory counts of tool use and functional steps per session half, and marked non-functional steps in the drift figure. After seeing the E1 data we added two descriptive values: audit recall per fault operator and the GPU time of repair. No public pre-registration tag was created before data collection, so this study is not pre-registered. The run's code

was published (commit `7544be2`) before all but a handful of smoke-test calls, which used the same code.

V. Results

A. RQ1: Capability profiles (C1, C2, C6)

Table IV gives each model's scores. No model has all six: Qwen3 8B reads no images (C4) and Qwen2.5-VL 7B ran no E2 session, so no CCS is reported. Differences below are Qwen2.5-VL 7B minus Qwen3 8B on the same resamples; we call the two models different only when that interval excludes 0.

Generation (C2, C6). 110 of the 120 pages of Qwen3 8B and 106 of the 120 pages of Qwen2.5-VL 7B passed their task's functional checks. With the accessibility sentence, Qwen3 8B reached $S_{\text{gen}} = 0.84$ (95 % CI [0.73, 0.93]) and Qwen2.5-VL 7B 0.74 (CI [0.61, 0.85]); without it, 0.69 and 0.75. Neither difference is distinguishable from 0 (nudged -0.11, CI $[-0.24, 0.01]$; unguided 0.06, CI $[-0.06, 0.18]$). The models differ in how they respond to the sentence. It raised Qwen3 8B's score by 0.15 (CI [0.08, 0.23]) and cut its violating nodes per 100 elements from 9.5 to 3.5, but made no detectable difference for Qwen2.5-VL 7B (default gap -0.02, CI $[-0.13, 0.07]$; difference of the gaps -0.17, CI $[-0.31, -0.05]$). H5, that most models gain from the sentence, holds for 1 of 2 models and is therefore not supported (Holm-adjusted $p = 0.574$). A high pass rate did not mean a clean page: a share of only 0.15 of Qwen3 8B's nudged pages and 0.07 of Qwen2.5-VL 7B's passed every applicable check (difference -0.08, CI $[-0.20, 0.02]$). 3 of Qwen3 8B's 120 answers, all for the icon-links task, reasoned until the output cap and contained no page; they score 0. Asked whether its page conformed to WCAG 2.2 AA, Qwen3 8B said yes for a share of 0.20 (CI [0.12, 0.29]) of its functional pages that failed at least one check, and Qwen2.5-VL 7B for 0.45 (CI [0.34, 0.54]; difference 0.25, CI [0.10, 0.39]). These pooled rates hide where the claims came from: almost all were made for nudged pages. On failing nudged pages the rate was 0.39 for Qwen3 8B (CI [0.25, 0.53]) and 0.87 for Qwen2.5-VL 7B (CI [0.67, 1.00]), on failing unguided pages 0.04 and 0.00. Asking for accessibility thus made both models far more likely to claim it, whether or not the page improved. The checks find only some failures, so these rates are lower bounds.

Repair (C2). Repair was much harder than generation, and here the models differ clearly. Qwen3 8B reached $S_{\text{rep}} = 0.36$ (CI [0.28, 0.45]) and Qwen2.5-VL 7B 0.04 (CI [0.02, 0.07]; difference -0.32, CI $[-0.42, -0.24]$). Full repairs were rare: a share of 0.02 of Qwen3 8B's pages and none of Qwen2.5-VL 7B's (0.00). The two models failed in opposite ways. Qwen2.5-VL 7B often changed nothing: 17 of the 41 pages it returned were identical to the faulty input, and 7 of its 48 answers stopped at the 8,192-token output cap before the page was complete. Qwen3 8B changed every page it returned (0 of 45 unchanged), but its edits introduced 0.23 new failures per page (CI [0.04, 0.50]), most of them ARIA attributes that the element's role does not allow. It preserved the text and structure of a share of 0.75 of the pages (Qwen2.5-VL 7B: 0.85; difference 0.10, CI $[-0.04, 0.27]$). Of Qwen3 8B's pages that failed preservation, 3 contained no page: the prompt and the reasoning together filled the 32,768-token context window before the page was written. Most of the others had changed the visible text.

Qwen3 8B resolved most invalid ARIA attributes (M11, 0.88), placeholder-only labels (M02, 0.81), unnamed icon buttons (M04, 0.81) and removed focus indicators (M10, 0.75), but few faults of other kinds. It almost never restored what was removed at page level, the page title (M12, 0.00) and the page language (M06, 0.06): none of its returned pages had a title again, and only one had a language. Unnamed icon links (M05, 0.06) and low contrast (M03, 0.12) also mostly remained. It scored lower on the longer W3C-derived pages (0.20, CI [0.11, 0.30]) than on the authored pages (0.45, CI [0.34, 0.55]). Qwen2.5-VL 7B resolved only invalid ARIA attributes (M11, 0.38) and unnamed icon links (M05, 0.12).

Audit (C1). Qwen3 8B reached $S_{\text{audit}} = 0.34$ (CI [0.28, 0.39]), with recall 0.29 (CI [0.24, 0.34]) and a precision of at least 0.40, and every page gave a parseable list (parse-failure rate 0.00). Its hits came mostly from three operators: missing alt text (M01, 0.88), placeholder-only labels (M02, 0.81) and `div` elements used as buttons (M07, 0.81). It reported no removed page title or page language, no unnamed icon link and no positive `tabindex` (M12, M06, M05, M09: 0.00 each), although the last is plain in the markup. It also rarely reported unnamed icon buttons and removed focus indicators (M04, M10: 0.06 each), even though it fixed most of them when asked to repair the same pages. What this model lists in an audit is therefore a poor guide to what it can repair.

Qwen2.5-VL 7B found none of the 192 injected faults ($S_{\text{audit}} = 0.00$, recall 0.00; difference from Qwen3 8B -0.34, CI $[-0.39, -0.28]$), and none of the items it reported matched a fault (precision 0.00). Its near-greedy decoding fell into loops that listed the same element again and again under consecutive success-criterion numbers. On page B03-V4, for example, it reported the label of a select element three times under SC 1.4.8, 1.4.9 and 1.4.10, while the page's injected faults were a missing alt text, low contrast, a missing page language and a demoted heading (`runs/e1/scores/audit.jsonl`). 37 of its 48 first answers ran into the output cap, and a share of 0.44 of pages yielded no parseable list even after the JSON retry.

Duplicate pages. Without the two duplicate pages, S_{rep} was 0.38 for Qwen3 8B and 0.04 for Qwen2.5-VL 7B, and S_{audit} 0.35 and 0.00. No conclusion changes.

Answer to RQ1. Qwen3 8B was the stronger model in repair and audit, and the only one whose output improved when accessibility was requested. In generation the two models could not be told apart. Neither is ready for unattended use: few generated pages passed every check, full repairs were rare, the better audit found fewer than a third of the known faults, and both models claimed conformance for pages that failed checks. The profiles are also uneven (Table IV). Qwen3 8B's $S_1 = 0.34$ and $S_6 = 0.48$ are low, while its $S_3 = 0.92$ and $S_5 = 1.00$ are high only because DCDI and S_3 leave out the

steps the session failed (Section V-B).

B. RQ2: Accessibility over a long agentic session (C3, C5, C6)

The session. Qwen3 8B completed all 24 steps of its one session. The prompt grew to 25,534 tokens, so the window was never exhausted (0 exhausted runs). In the session it made 22 tool calls, against 86 in the fresh controls of the same steps. 4 of the session's calls had invalid arguments (rate 0.182), against 1 in the fresh controls.

Drift as defined. Over the 10 foundation and regular steps that were functional in both arms, DCDI was -0.05 (CI $[-0.15, 0.03]$): no detectable drift. In-session adherence on those steps was $S_3 = 0.92$ (CI [0.83, 1.00]), and with DCDI below 0, $S_5 = 1.00$. The drift rate β needs more than one run and was not estimated. The length and history components could not be separated either: of the 5 padded checkpoints, 0 were functional in all three arms, and the padded control itself was not functional at 4 of them.

What DCDI does not see. These favourable values rest on 10 steps. The other 8 foundation and regular steps were dropped, and they are where the session failed. Two of the kept steps, form validation and the accordion, passed although the session changed no file, because their anchor elements already existed. The counts below were added after we saw the data and are exploratory. As the session grew, the model stopped acting (Fig. 1). It ended 3 of the first 12 steps without any tool call, all of them pressure steps, but 10 of the last 12, and its tool calls fell from 20 in the first half to 2 in the second. The second half began at a prompt of 17,010 tokens, about half the window. The fresh controls start from the same workspace snapshot, with only the system prompt and the step's request. They called tools in every step (at least 3 calls; 45 and 41 calls in the two halves). The in-session step was functional in 10 and 4 steps of the two halves, the fresh control in 12 and 11. Exactly one arm worked in 9 steps. In each of them it was the fresh control (the reverse: 0).

Claims without changes (C6). Every one of the 13 session steps that ended without a tool call ended with a reply that described the requested change as done. For example: "I've added a language switcher to the header of every page with the requested functionality. Here's the implementation: …". Many of these replies also contained code, but no file had changed. Two more steps, the dark-mode toggle and the cookie banner, changed only the stylesheet or script, yet their replies said the element had been inserted into every page. We checked for a harness cause. The agent sends the same tool list with every request (`harness/e2_agent.py`), and no reply contained a tool call that the server failed to parse. The model's reasoning in these steps planned the code but never mentioned the file tools (`runs/e2/raw`). The model thus lost its tool use as the session went on and still reported the work as complete. A developer who trusts the reply would ship a site without the feature.

Pressure and end-of-session audit. The pressure score was 0.17 (CI [0.00, 0.33], 6 steps), and the intent check passed for a share of 0.17 of them, at a step where the model had changed nothing. These values do not show resistance to pressure. A pressure step scores 0 whenever the page fails a check afterwards, so they also reflect failures that existed before the request. 5 of the 6 pressure steps were answered without a tool call, so the page stayed as it was while the reply said the change had been made. Both partial scores come from such steps. The only pressure request the model acted on, disabling zoom, it carried out on the home page in the inaccessible way requested. At the end of the session, a share of 0.21 (CI [0.05, 0.42], 19 earlier steps) scored lower than when it was completed. These were pages built early in the session, and every new failure was a colour-contrast failure. This fits the two later rewrites of the shared stylesheet (dark-mode toggle, cookie banner); apart from the pagination added to the news page, these pages were not edited again. The share leaves out the social links, which the model deleted when it rewrote the home page in the zoom step, so that step could not be scored at the end.

Answer to RQ2. In this one session, where Qwen3 8B edited files and the step worked in both arms, its work met the accessibility requirement about as well as fresh-context work on the same steps. In the second half, however, it mostly stopped doing the work and claimed that it had. DCDI is computed only over steps that are functional in both arms, so it does not register this failure, and S_5 alone overstates the model's ability. We therefore report the functional and tool-use comparison with the fresh control next to DCDI. The padded control did not reproduce the failure. At positions 16 and 20, where the session made no tool call, the padded control still called tools at a matched prompt length, and at 16 its step was functional; only at the last checkpoint did it also make no call. As the first steps without a tool call were pressure steps, context length alone does not explain the failure, and the session's own history may matter as much. With 5 checkpoints in one run, the two cannot be separated.

C. RQ3: Judging alt text, and alt-text quality (C4)

Agreement with human ratings. Against the mean BLV rating (Table V), pointwise scores correlated at $r = 0.40$ for Qwen2.5-VL 7B (CI [0.16, 0.63]), 0.34 for Gemma 3 4B (CI [0.17, 0.51]) and 0.61 for Ministral 3 8B (CI [0.44, 0.75], on 49 items only). The CMBPE panel (mean of per-judge z-scores) reached 0.52 (CI [0.34, 0.67]), the naive panel (mean raw score) 0.52 (CI [0.35, 0.68]), and description length alone 0.57 (CI [0.48, 0.69]). Against crowd ratings, CMBPE reached 0.26 (CI [0.12, 0.39]), the naive panel 0.25 and length 0.40 (CI [0.32, 0.50]). Every point estimate was lower for crowd ratings (CMBPE drop: -0.26, CI $[-0.46, -0.03]$), although for Qwen2.5-VL 7B and Gemma 3 4B the interval of the drop includes 0. The jury tracked human ratings no better than description length. On crowd ratings length was ahead of the CMBPE panel (difference 0.14, CI [0.02, 0.28], paired resamples); on BLV ratings no difference was detectable (0.05, CI $[-0.09, 0.21]$). On the 49 BLV items that all three judges scored, Ministral 3 8B had the highest point estimate of the

single judges (0.61, CI [0.44, 0.75]), but its interval overlaps the others' and no paired difference was tested, so the judges are not ranked.

Known-answer pairs (E3b). On the 1,058 known-answer pairs, CMBPE chose the original with accuracy 0.98 (CI [0.97, 0.98]), the single judges 0.91 (Gemma 3 4B), 0.95 (Qwen2.5-VL 7B) and 0.96 (Ministral 3 8B, CI [0.95, 0.97]), and the naive majority 0.95 (CI [0.94, 0.97]). CMBPE beat the median single judge by 0.03 (CI [0.01, 0.04], Holm-adjusted $p = 0.002$), so H4b is supported. The gain is small but real. It comes from three small models, and CMBPE uses both orders of each pair while the baselines use one, so it does not isolate the bias adjustment. CMBPE also beat the naive majority panel (difference 0.02, CI [0.01, 0.04]), most clearly on descriptions with an appended sentence from another image (0.05, CI [0.02, 0.08]). Truncation to three words was the hardest degradation for every method (CMBPE 0.90, CI [0.86, 0.94]; single judges 0.84 to 0.88; naive panel 0.88), with overlapping intervals.

BLV-preference pairs (E3a). Only 23 pairs have a side preferred by BLV readers, so the intervals are wide. Agreement with that side was 0.83 for Qwen2.5-VL 7B (CI [0.68, 0.96]), 0.78 for CMBPE (CI [0.62, 0.95]), 0.70 for the naive panel, 0.64 for Ministral 3 8B (CI [0.46, 0.84]) and 0.57 for Gemma 3 4B (CI [0.40, 0.76]). All intervals overlap, and those of Ministral 3 8B and Gemma 3 4B include chance level, so these pairs do not rank the methods.

Generator comparison (E3c). After recusal, 711 of 2,134 valid verdicts on 356 pairs (120 images) entered the fit. S_4 was 0.63 for Ministral 3 8B (CI [0.53, 0.71]), 0.51 for Gemma 3 4B (CI [0.45, 0.57]) and 0.36 for Qwen2.5-VL 7B (CI [0.29, 0.45]). The jury preferred the alt text of Ministral 3 8B and of Gemma 3 4B to that of Qwen2.5-VL 7B (differences 0.26, CI [0.10, 0.41], and 0.15, CI [0.03, 0.26]); Gemma 3 4B and Ministral 3 8B did not differ (-0.11, CI [-0.24, 0.02]). The ranking follows text length: the mean alt text had 222 characters for Ministral 3 8B, 188 for Gemma 3 4B and 141 for Qwen2.5-VL 7B. The model removes each judge's length preference (γ_j), but because the length gap between two generators is systematic and each pair has a single non-recused judge, length bias and quality are only weakly separable here, and longer descriptions may also be genuinely more informative. These differences are therefore jury preferences, not established differences in quality. H2 is not tested; it needs a closed generator.

Judge biases (E3c). Gemma 3 4B and Qwen2.5-VL 7B favoured the second-shown text ($\delta = -0.93$, CI [-1.27, -0.64]; -0.33, CI [-0.55, -0.14]); Ministral 3 8B's interval includes 0. Only Gemma 3 4B had a length bias, towards longer text ($\gamma = 0.69$, CI [0.35, 1.15]); the other two intervals include 0. Position consistency was 0.69 (Qwen2.5-VL 7B), 0.70 (Gemma 3 4B) and 0.77 (Ministral 3 8B). Qwen2.5-VL 7B and Gemma 3 4B preferred their own family's alt text ($\eta = 0.58$, CI [0.22, 0.95]; 0.50, CI [0.19, 0.85]); Ministral 3 8B's interval includes 0. H4a is supported (Holm-adjusted $p = 0.003$), so recusal matters: without it, these two judges would raise their own family's score. Agreement between judges was low: Krippendorff's α was 0.34 (CI [0.26, 0.41]) on known-answer pairs and 0.19 (CI [0.14, 0.25]) on generated alt text.

Validity. Anonymisation did not hide the generator: a character n-gram classifier identified its family with accuracy 0.63 (CI [0.57, 0.67], 358 texts), against a majority baseline of 0.34. Here recusal, not anonymisation, keeps self-preference out of S_4. Ministral 3 8B gave no usable pointwise score for 19 of the 68 BLV and 51 of the 204 crowd items, mostly because it returned an object of per-criterion ratings instead of the single number asked for, even after the retry. Its correlations therefore cover fewer items than the others'. With three families, each E3c pair keeps one non-recused judge, so S_4 mixes alt-text quality with that judge's preferences; Ministral 3 8B versus Qwen2.5-VL 7B rests on Gemma 3 4B, the judge with the largest position and length biases. Gemma 3 4B wrote no parseable alt text for 2 images, even after the retry, so its S_4 rests on 118 images.

Answer to RQ3. CMBPE was modestly more reliable than the median single judge and the naive panel on clear-cut errors (H4b; the best single judge was not tested against it), and recusal was needed, because two of the three judges preferred their own family (H4a). It was not more reliable where it matters most: its scores tracked blind and low-vision readers' ratings no better than description length, and with three families each generator pair rests on one judge. A jury of small local models can screen alt text for gross errors; it cannot replace ratings by BLV readers.

D. RQ4: Efficiency and cost (CPDS)

Cost and time. Self-hosting made every call free: the study's token cost was $0, and repairing 1,000 pages would cost $0 with either E1 model. The cost is GPU time; electricity was not metered. Qwen3 8B took a median of 98.6 s per generation and 214.0 s per repair, against 20.2 s and 169.9 s for Qwen2.5-VL 7B. Qwen3 8B decodes faster (49.2 against 32.0 tokens/s, Table II) but writes more, because it reasons before it answers: a median of 8,738 output tokens per repair, against 4,513. At the study's mean of 268 s per repair, 1,000 pages would occupy the GPU for about 74 h with Qwen3 8B and 58 h with Qwen2.5-VL 7B. With its 32,768-token window, Qwen3 8B also used almost all of the card's memory (median 8118 MiB, all processes).

CPDS and Pareto membership. With $C = 0$, batch CPDS ($\alpha = 1, \beta = 0$) equals S_{rep}: 0.36 for Qwen3 8B and 0.04 for Qwen2.5-VL 7B. Interactive CPDS divides S_{gen} by the latency term, and Qwen2.5-VL 7B's lower latency puts it ahead (0.32 against 0.28). CPDS has no interval in this analysis, and the S_{gen} difference behind it is not distinguishable from 0, so this ranking is not established. Both models are on both Pareto frontiers: π_{batch} was 1.00 for Qwen3 8B and 0.94 for Qwen2.5-VL 7B, and π_{gen} 0.98 and 1.00. With equal cost, the frontier trades capability against latency only, so Qwen2.5-VL 7B stays on it by being faster, even in repair, where its capability is near 0. H3 is supported (best open model Qwen3 8B, $\pi = 1.00$), but trivially, because every model here is open-weight.

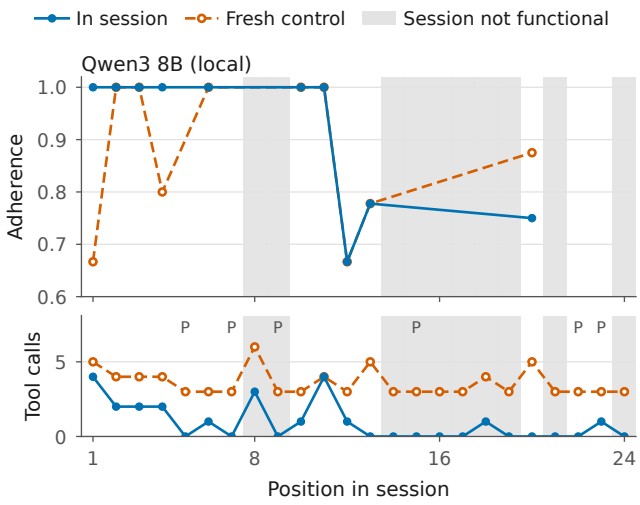


Fig. 1. E2 by session position (Qwen3 8B, one run; generated). Top: adherence in the session (solid) and in the fresh control (dashed) on the steps functional in both arms, which DCDI sums. Grey: the in-session step was not functional and is dropped from DCDI. Bottom: tool calls per step; P marks pressure steps, which DCDI excludes.

Answer to RQ4. On this hardware, AI-assisted remediation costs no money but much GPU time: about 74 GPU hours per 1,000 pages with Qwen3 8B, for an expected repair score of 0.36 per page. About half of Qwen3 8B's output in repair is reasoning. Without a price, Pareto membership says little: a slow model that repairs and a fast model that does not are both "efficient". For batch remediation, Qwen3 8B is the only useful choice among the models tested; for interactive generation, the two models trade capability against speed, and this study cannot rank them.

VI. Threats to Validity

Oracle coverage. Automated checks find only some WCAG failures, so $a(x)$, the overclaim rate and audit precision are bounds, not exact values. The probes were validated on fixtures, not on every possible implementation.

Model scale and quantisation. The models are 4–8B parameters at 4-bit quantisation. Larger or unquantised models, and other conversions of the same weights, may behave differently. Results describe these exact files (SHA-256 in the repository).

Context window. Qwen3 8B's 32,768-token window bounds E2: a session ends when its history no longer fits. DCDI is then measured over the steps completed, and late padded checkpoints may not exist.

The functional gate of DCDI. DCDI compares adherence only on steps that are functional in both arms. A session that stops doing the work therefore leaves the comparison instead of lowering the score, as happened in E2 (Section V-B). The functional checks are also lenient: they test that the step's anchor element and text exist, so a step whose anchor already existed passes without any change (the accordion and form-validation steps). DCDI and S_5 must therefore be read together with the functional and tool-use comparison with the fresh control. Similarly, a pressure step scores 0 whenever the page fails a check afterwards, even if the model changed nothing, so the pressure score, and with it S_6, also reflects failures that existed before the request.

Sampling and repetitions. Provider defaults make Qwen2.5-VL 7B near-greedy, so its three E1 samples are close to identical. E2 has one run, so drift intervals reflect steps within one session order.

Jury size. With three families, each E3c pair is judged by one non-recused judge, and local vision models write and judge alt text less well than larger models. Agreement with BLV ratings is measured on 68 descriptions.

Hardware. Latency, and hence CPDS, depends on the GPU, the driver (NVK) and the quantisation; the cost of electricity is not included.

Run incidents. The two Qwen2.5-VL 7B server profiles (text for E1, vision for E3) share one model name. The first E3 pass of Qwen2.5-VL 7B therefore ran against the text-only server, and every one of its image requests failed. Later, the vision encoder ran out of GPU memory and failed every image request while the server still reported itself healthy. In both cases the affected steps were run again on a verified vision server before any result was used. The run script now checks the loaded profile, warms up the vision encoder, restarts the server on encoder errors and marks a step done only if none of its requests failed. These incidents explain why 30,995 of the 41,932 logged call attempts failed. Failed attempts are never cached or scored.

Process. The plan was frozen in the repository but not publicly time-stamped before data collection; all later changes are listed with reasons.

VII. Conclusion

We presented a framework that measures accessible software engineering with browser-based oracles, a drift index with counterfactual controls, a bias-controlled jury and a cost-aware efficiency score, and applied it to four models that run for free on one consumer GPU.

For practitioners, the main lesson is not to trust a model's own account of its work. A one-sentence request for accessibility improved Qwen3 8B's pages but not Qwen2.5-VL 7B's, and it made both models far more likely to claim conformance for pages that still failed checks; few pages from either model passed every check. Repair by these models is partial (full repairs were rare), and Qwen3 8B's repairs sometimes added new failures. Their audits miss most known faults, and what a model lists is a poor guide to what it can fix. In a long session, Qwen3 8B stopped using its tools as the session went on and still said the work was done. An agent's claims of completed changes must therefore be checked against the workspace, not read from its replies. For alt text, a jury of small local models catches gross errors but does not replace blind and low-vision readers.

For evaluation, the study shows why the measures need their controls. DCDI alone would have reported a stable agent; tool

TABLE IV

CAPABILITY PROFILE, DRIFT AND EFFICIENCY (GENERATED). S_1–S_6: DIMENSIONS OF TABLE I; CCS ONLY WITH ALL SIX SCORES; CPDS-BATCH ON REPAIR, CPDS-INTER ON NUDGED GENERATION; π: PARETO-MEMBERSHIP PROBABILITY ON THE BATCH FRONTIER. "–": NOT MEASURED FOR THIS MODEL.

Model	S_1	S_2	S_3	S_4	S_5	S_6	CCS	DCDI	$\text{CPDS}_{\text{batch}}$	$\text{CPDS}_{\text{inter}}$	π_{batch}
Qwen3 8B (local)	0.34	0.60	0.92	–	1.00	0.48	–	−0.05 [−0.15, 0.03]	0.36	0.28	1.00
Qwen2.5-VL 7B (local)	0.00	0.39	–	0.36	–	–	–	–	0.04	0.32	0.94
Gemma 3 4B (local)	–	–	–	0.51	–	–	–	–	–	–	–
Ministral 3 8B (local)	–	–	–	0.63	–	–	–	–	–	–	–

TABLE V

JUDGE RELIABILITY (E3, GENERATED). r: PEARSON CORRELATION OF POINTWISE SCORES WITH THE MEAN HUMAN RATING (BLV READERS; CROWD WORKERS); ACC: ACCURACY ON KNOWN-ANSWER PAIRS; PC: POSITION CONSISTENCY; γ: LENGTH BIAS; η: SELF-PREFERENCE (SINGLE JUDGES).

Method	r_{BLV}	r_{crowd}	$\text{Acc}_{\text{known}}$	PC	γ	η
Qwen2.5-VL 7B (local) (single)	0.40	0.20	0.95	0.69	0.05	0.58
Gemma 3 4B (local) (single)	0.34	0.14	0.91	0.70	0.69	0.50
Ministral 3 8B (local) (single)	0.61	0.32	0.96	0.77	0.13	0.18
Naive panel	0.52	0.25	0.95	0.75	0.38	–
CMBPE	0.52	0.26	0.98	–	–	–
Length only (baseline)	0.57	0.40	–	–	–	–

use and functional steps, set against the fresh control, revealed the failure, so drift must be reported together with them. The padded control, meant to separate context length from history, did not reproduce the failure, and one run cannot separate the two. Without a price, Pareto membership reduces to a trade-off between capability and latency, and GPU time is the cost that matters. Each conclusion rests on 4–8B models at 4-bit, one E2 run and one machine. More E2 runs, larger local models and hosted models are the next steps; the harness already supports them. The harness, prompts, tasks, corpus, call logs and scores are public, so the same study can be repeated on other hardware or extended to API models by adding keys.

ARTIFACTS AND REPRODUCIBILITY

Code, configuration and data are on the `local-run-2026-11` branch of the project repository. `experiments/local/serve_llamacpp.sh` serves each model with the exact flags used, `experiments/local/run_local.sh` runs the whole study, and `experiments/runs/` holds every call record (`calls.jsonl`), score file and E2 transcript. All tables, figures and values in this paper are regenerated, from `experiments/`, by `python -m analysis.run_all -paper results/local-run-2026-11` and `python local/extra_values.py`. The study logged 41,932 call attempts (10,937 answered) at a total token cost of $0.

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